

2021-Ph-H1-Q7

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Two trolleys stick together in a collision.

Find their final speed.

Solution:

Use conservation of momentum:

$$(2 \times 3) + (1 \times 0) = (2 + 1)v \Rightarrow v = \frac{6}{3} = 2 \text{ m/s}$$

Answer: D

Guidance for Students:

Always apply momentum conservation **before and after** the collision. Sticky = inelastic, but momentum still conserved.

Revision Tip:

Momentum is always conserved in collisions. Use:

$$m_1v_1 + m_2v_2 = (m_1 + m_2)v$$